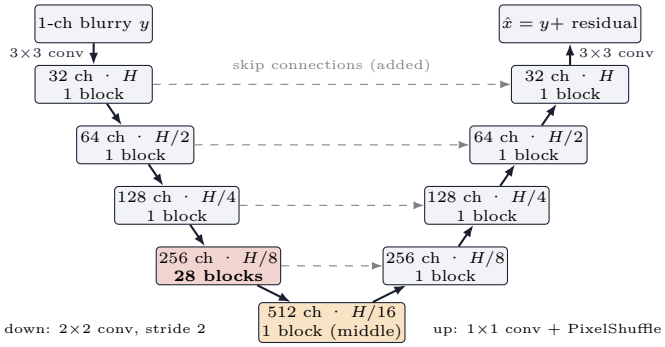
Has NAFNet (M1) overfit? A one-page answer

Short answer. NAFNet has **not** memorised its training images: it scores the same on images it trained on as on images it never saw (train−test gap +0.04 dB beyond a control). It **has** overfit the *kinds of blur* it was trained on: on unseen motion directions it loses 3.6 dB, ending only about 2 dB above the blurred input. That is a distribution-shift failure, not classic overfitting, and more images would not fix it.

1. Architecture (NAFNet, width 32, 17.1 M parameters)

A U-Net of 36 *NAFBlocks*. The network predicts a residual that is added to the blurred input; it never receives the blur kernel k or the noise level σ.



Inside a NAFBlock (no ReLU/GELU anywhere): LayerNorm → 1×1 conv → 3×3 depthwise conv → *SimpleGate* (split channels, multiply halves) → simplified channel attention (global average pool + 1×1 conv) → 1×1 conv → residual add; then LayerNorm → 1×1 conv → SimpleGate → 1×1 conv → residual add. Residual branches start scaled by zero, so each block starts as the identity.

Layer type	Count	Where
NAFBlocks	36	enc. 1+1+1+28, middle 1, dec. 1+1+1+1
Conv layers, total	226	
1×1 conv	184	5 per block + 4 upsamplers
3×3 depthwise conv	36	1 per block
3×3 conv	2	intro, ending
2×2 stride-2 conv	4	downsampling
LayerNorm / SimpleGate	72 / 72	2 each per block
Channel attention	36	1 per block
PixelShuffle	4	upsampling
Parameters	17.1 M	76% in the 28 deepest blocks

2. Training

One model for all 12 training kernels (P_1: 4 Gaussian, 8 linear motion at 0/45/90/135°) and all 5 noise levels. AdamW, L2 loss, batch 16, learning rate 2×10^{-4} with 1k warm-up and cosine decay, gradient clipping 1.0, bfloat16, 100,000 iterations (2.8 h on an RTX 3090 Ti). Checkpoint = best mean PSNR on a fixed validation set; the test set is never used for selection.

3. How 200 images become 1.6 million inputs

Inputs are generated on the fly. For every training example:

1. pick one of the **200** training images (481×321);
2. cut a random **128×128 crop**: 354×194 = **68,676** possible positions;
3. random 90° rotation and flip: **8** variants;
4. blur with a random training kernel: **12** choices;
5. add Gaussian noise at a random level: **5** levels, with **fresh random noise** every time.

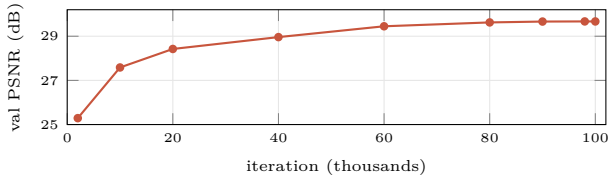
16 patches × 100,000 iterations = 1,600,000 training inputs (26.2 billion pixels): about **8,000 different patches per image** and 26,700 per (kernel, noise) pair. Before noise there are $200 \times 68{,}676 \times 8 \times 12 \times 5 \approx 6.6 \times 10^9$ distinct inputs, and the noise is never the same twice, so the network practically **never sees the same input twice**.

4. Train, validation and test results

PSNR, dB (gain vs. blurred)	NAFNet	ADMM-EPLL†
Train: 20 train images, P_1	27.45 (+5.48)	27.33 (+5.36)
Test: 20 test images, P_1	27.75 (+5.52)	27.67 (+5.44)
Train − test gap in gain	−0.04	−0.08
Validation: 10 images	29.67	–
Full test, 200 images, P_1	27.48 (+5.38)	27.43 (+5.33)
P_2-interp (unseen settings)	24.57 (+2.73)	27.08 (+5.24)
P_2-extrap (unseen defocus)	23.28 (+2.49)	25.38 (+4.60)

†Physics-aware classical solver (knows k, σ); its only learned part is a patch prior, so it is the control for how hard each image set is. Train/test: $\sigma > 0$, identical cells. U-Net shows the same picture (train − test gap −0.14 dB). Validation includes $\sigma = 0$, hence the higher number.

Validation PSNR during training: rises, then flattens; it never turns down



5. Verdict

Not overfit: the training images. Train and test scores match (gap +0.04 dB beyond the control, Mann–Whitney $p = 1.0$), and validation PSNR peaks at 98k of 100k iterations without ever declining.

Overfit: the training kernels. The gain over the blurred input halves on unseen kernels (+5.38 → +2.73 / +2.49 dB), and on unseen motion directions NAFNet keeps only about 2 dB. The physics-aware solver barely changes (+5.33 → +5.24 / +4.60 dB).

Data-hungry? Not tested. More iterations would not help (flat curve), and NAFNet beats the smaller U-Net by 0.7 dB on the same data. A learning curve (retrain on 50 and 100 images, about 6 h GPU) would settle it.